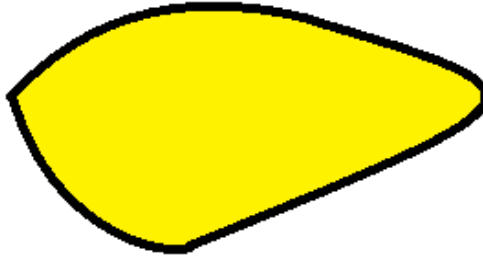


1.7 Centers of Gravity

General Idea

Suppose we have a flat plate (called a *lamina*) of some material with uniform density which occupies a region of the xy -plane. A picture of a possible “plate” (we will assume plates are two dimensional) is below:



Intuitively speaking the center of mass/gravity of the plate (also called the centroid) is the point at which the plate will balance perfectly.

If a plate with uniform density, ρ , occupies the region in the xy -plane bounded by a curve given by $y = f(x)$ where $a \leq x \leq b$ and the x -axis, then the mass of the plate is given by $m = \rho \int_a^b f(x) dx$ provided that $f(x)$ is a continuous nonnegative function.

Note: The formula for mass should remind you of density times volume. Only in this case we use density times area since a lamina is two dimensional.

Formula (Moments and Center of Gravity/Mass): Suppose $f(x)$ is a continuous nonnegative function. If a plate with uniform density, ρ , occupies the region in the xy -plane bounded by the curve given by $y = f(x)$ where $a \leq x \leq b$ and the x -axis, then:

- The moment of the plate about the y -axis is $M_y = \rho \int_a^b x f(x) dx$.
- The moment of the plate about the x -axis is $M_x = \rho \int_a^b \frac{1}{2} (f(x))^2 dx$.
- The center of gravity (centroid) of the plate is $(\bar{x}, \bar{y})$ where $\bar{x} = \frac{M_y}{m} = \frac{1}{A} \int_a^b x f(x) dx$ and $\bar{y} = \frac{M_x}{m} = \frac{1}{A} \int_a^b \frac{1}{2} (f(x))^2 dx$ where A is the area of the region that the plate occupies.

Note: As we can see in the above formulas the center of gravity does not depend on the density when the density is uniform. In Calculus 3 you will see (as you would probably expect) that the center of gravity does depend on the density when the density is not uniform.

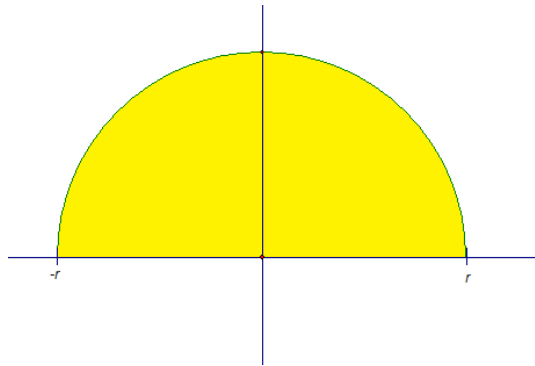
Note: The book present reasoning behind these formulas at the start of the section. I have decided to omit this from the notes, since you will return to this problem (in much more generality) in Calculus 3.

Note: Intuitively speaking we can view (as the notation suggests) $\bar{x}$ and $\bar{y}$ as the average x and y coordinates respectively in the region which the plate occupies.

Example 1: Find the center of gravity of the plate with uniform density which occupies the region:

$$R = \left\{ (x, y) \mid -r \leq x \leq r \text{ and } 0 \leq y \leq \sqrt{r^2 - x^2} \right\} \text{ where } r \text{ is some positive real number.}$$

A picture of the region is:



Using our knowledge of area of a circle we have the area of the region is $\frac{\pi r^2}{2}$. Now, by the above

$$\text{formulas } \bar{x} = \frac{2}{\pi r^2} \int_{-r}^r x \sqrt{r^2 - x^2} dx = \frac{2}{\pi r^2} \left(\frac{-1}{3} (r^2 - x^2)^{\frac{3}{2}} \right) \Big|_{-r}^r = 0 \text{ and}$$

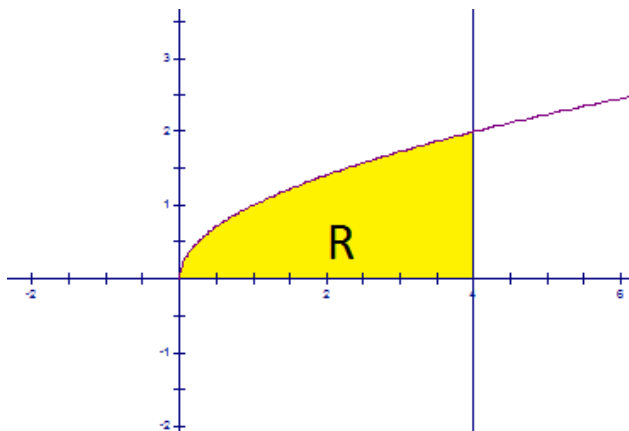
$$\begin{aligned} \bar{y} &= \frac{2}{\pi r^2} \int_{-r}^r \frac{1}{2} (\sqrt{r^2 - x^2})^2 dx = \frac{1}{\pi r^2} \int_{-r}^r (r^2 - x^2) dx = \frac{1}{\pi r^2} \left(\left(r^2 x - \frac{1}{3} x^3 \right) \Big|_{-r}^r \right) \\ &= \frac{1}{\pi r^2} \left(r^3 - \frac{r^3}{3} - \left(-r^3 + \frac{-r^3}{3} \right) \right) = \frac{1}{\pi r^2} \left(\frac{4r^3}{3} \right) = \frac{4r}{3\pi}. \end{aligned}$$

Thus, the center of gravity (the centroid) is $\left(0, \frac{4r}{3\pi} \right)$.

Question: Does this answer make sense intuitively?

Example 2:

- a) Sketch the region, R , in the first quadrant bounded by the curves $y = \sqrt{x}$, $y = 0$, and $x = 4$.



- b) Suppose a plate with uniform density occupies the region R . Find the exact center of gravity of the plate.

First, we calculate the area of the plate. It is $\int_0^4 \sqrt{x} \, dx = \frac{2}{3} x^{\frac{3}{2}} \Big|_0^4 = \frac{16}{3}$. By the given formulas

$$\bar{x} = \frac{3}{16} \int_0^4 x \sqrt{x} \, dx = \frac{3}{16} \left(\frac{2}{5} x^{\frac{5}{2}} \Big|_0^4 \right) = \frac{3}{40} (32) = \frac{12}{5} \quad \text{and} \quad \bar{y} = \frac{3}{16} \int_0^4 \frac{1}{2} (\sqrt{x})^2 \, dx = \frac{3}{32} \int_0^4 x \, dx = \frac{3}{32} \left(\left(\frac{1}{2} x^2 \right) \Big|_0^4 \right) = \frac{3}{4}.$$

Thus, the center of gravity (the centroid) is $\left(\frac{12}{5}, \frac{3}{4} \right)$.